

Building and Evaluating a Regime-Adaptive Trading Framework for Indian Equities

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JEL Classification: G11, G14, G17, C63

Keywords: algorithmic trading, systematic trading, Indian equities, technical analysis, risk management, backtesting, market regime detection, momentum strategies, mean reversion, position sizing

Suggested Citation: Prakhar, V. (2026). *Building and Evaluating a Regime-Adaptive Trading Framework for Indian Equities. Working Paper. Available at SSRN.*

Abstract

This paper documents the design, implementation, and back-tested evaluation of a modular algorithmic trading system developed for Indian large-cap equities listed on the National Stock Exchange (NSE). The framework integrates a technical indicator computation engine, a rule-based market regime classifier, three distinct trading strategy modules — trend following, momentum, and mean reversion — and a risk management layer built around fixed-fractional position sizing and stop-loss enforcement. The system is evaluated over a three years back-testing window spanning January 2021 through January 2024 across five constituent equities of the Nifty 50 index.

A distinguishing feature of the architecture is the regime-adaptive strategy selection mechanism, which classifies prevailing market conditions into one of five states — trending upward, trending downward, ranging, volatile, or calm — and dynamically routes capital to the most appropriate strategy. Position sizing follows the fixed-fractional risk model wherein no more than two percent of portfolio value is placed at risk per trade, with a supplementary cap of twenty percent of capital per position. Stop-loss conditions are evaluated daily against intraday low prices rather than closing prices, offering a more realistic simulation of execution in volatile sessions.

Across the January 2021–January 2024 back-testing window, four of five symbols produced positive nominal returns (ranging from +3.54% to +7.65%), with HDFCBANK.NS the lone exception at -3.31% — a result tied closely to its difficult 2022–2023 merger period. All five underperform the Nifty 50 buy-and-hold benchmark, which returned roughly 55% over the same span. That gap is expected and worth understanding rather than apologizing for: a system designed around signal confirmation and stop-loss discipline will miss large parts of a sustained rally. Maximum draw-downs stayed below 6.1% across all symbols. Sharpe ratios are negative relative to India’s 6.5% risk-free rate — a point the paper engages with directly rather than glosses over. The results are discussed in Section 6 in terms of what they reveal about the system’s behaviour, not just its performance.

To be clear about scope upfront: this is a three-year back-test on five stocks, using a rule-based regime classifier, with no walk-forward validation. Those are real constraints, not false modesty. The framework is not positioned as a production strategy — it is positioned as a documented, reproducible system that independent researchers can extend. The limitations are discussed in detail in Section 7, and they are worth reading before drawing conclusions from the results.

1. Introduction

1.1 The Rise of Retail Systematic Trading

Over the past decade, the barrier to entry for systematic trading has fallen substantially. The combination of commission-free data APIs, accessible programming environments, and growing educational resources has enabled a generation of technically literate retail participants to construct, test, and deploy automated trading strategies with infrastructure that, not long ago, would have required institutional-scale resources. India presents a particularly interesting setting for this development: the country's retail investor base expanded dramatically during and after the COVID-19 pandemic, with the number of demat accounts crossing 100 million for the first time in 2023 (SEBI Annual Report, 2023). Alongside this growth, there has been a measurable uptick in retail interest in algorithmic approaches, driven partly by the proliferation of API-enabled brokers such as Zerodha, Upstox, and Angel One.

Despite the accessibility of tools, the gap between a working prototype and a rigorously evaluated systematic strategy remains wide. Many retail algorithms are built without well-specified risk management, rely on back-testing methodologies that overestimate performance through look-ahead bias or unrealistic execution assumptions, and are designed around a single strategy that is implicitly assumed to work across all market conditions. This paper documents an attempt to address these shortcomings within the constraints of a retail-scale research project.

1.2 Problem Statement and Motivation

The central question motivating this work is straightforward: can a rule-based, modular trading framework — one that adapts its strategy selection to prevailing market conditions, enforces disciplined position sizing, and applies realistic execution assumptions — generate defensible risk-adjusted returns on Indian large-cap equities? The emphasis here is on defensible. The system is not designed to maximize raw returns, but to provide a systematic and explainable basis for trade decisions, with downside protection embedded at each layer.

A secondary motivation is architectural. Much of the published literature on retail algorithmic trading is either highly theoretical (deriving optimal strategies under stylised assumptions) or entirely empirical without transparent methodology. This paper attempts an intermediate position: a fully specified, reproducible implementation with honest acknowledgment of its assumptions and failure modes.

1.3 Research Objectives

This paper has four practical objectives. First, the system is documented thoroughly enough that someone with intermediate Python skills and access to NSE data could rebuild it from scratch — every design decision is explained, not just described. Second, each component choice is

grounded in existing literature rather than presented as obvious. Third, the results are discussed honestly: what a three-year back-test on five stocks can and cannot tell you is a short list, and the paper tries to stay within it. Fourth, the limitations section is written to be useful — pointing to specific, actionable extensions rather than the usual boilerplate about future work.

1.4 Contributions

Three contributions are worth highlighting. The back-testing loop evaluates stop-loss conditions against intraday lows before processing any strategy signals — a small but meaningful difference from close-price-only simulations that underestimate stop-loss frequency. The regime-strategy mapping provides a simple mechanism to avoid running momentum signals in volatile markets or trend signals in sideways ones; simple as it is, this kind of routing logic is often absent in retail back-testing implementations. And while the empirical scope is narrow, documented back-testing studies of regime-adaptive systems on NSE large-caps from outside institutional settings are not common — there is at least some value in adding one to the public record.

2. Literature Review

The academic debate on technical analysis centers on its compatibility with market efficiency. Fama (1970) argued that price-based rules cannot generate excess returns in an efficient market; subsequent work, including Lo, Mamaysky, and Wang (2000), provided empirical evidence of statistically significant patterns in technical indicators under specific conditions. In the Indian context, studies have documented short-run momentum effects and questioned weak-form efficiency in NSE data, providing a partial theoretical basis for the indicator-based approach used here.

The use of multiple confirmation indicators to reduce false signals is discussed in Chan (2009) and Aronson (2007), both of whom note that composite entry conditions reduce signal frequency while improving signal quality. The regime detection framework draws on Hamilton's (1989) Markov-switching model and practical extensions by Ang and Bekaert (2002), though the implementation here uses a deterministic rule-based classifier rather than a statistical model — a deliberate simplification that prioritizes interpretability over statistical precision.

On risk management, Van Tharp (1997) and Vince (1992) both identify position sizing as a more important determinant of long-run performance than entry signal quality. The fixed-fractional sizing model used here — risking two percent of capital per trade — follows standard practitioner guidance. Bailey, Borwein, López de Prado, and Zhu (2014) provide the definitive treatment of over-fitting in back-testing; the present work does not claim to circumvent these problems, and the parameter choices are documented explicitly to enable reproducibility and future stress-testing.

3. System Architecture

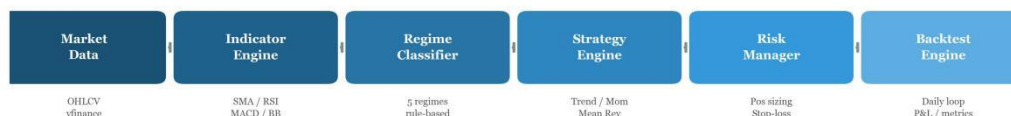
3.1 Modular Design Philosophy

The system is organized around six loosely coupled modules: data ingestion, indicator computation, regime classification, strategy execution, risk management, and back-testing simulation. Each module exposes a clean interface and can be modified or replaced independently, which is important both for testing and for eventual deployment. The main execution entry point orchestrates the pipeline sequentially for each symbol in the configured universe.

Figure 1 provides a schematic overview of the pipeline. Data flows from the market data fetcher through the indicator engine, at which point the enriched data-frame is passed to the

regime classification and strategy selection module. The selected strategy generates a signal column (1 for buy, -1 for sell, 0 for hold) which is consumed by the back-test engine. The risk manager is invoked at two points: at entry to calculate position size, and daily throughout the holding period to evaluate stop-loss conditions.

Figure 1: System Architecture of the Regime-Adaptive Trading Framework



Data flows left-to-right through six loosely coupled modules. Each module exposes a clean interface and can be modified independently.

Figure 1: System Architecture of the Regime-Adaptive Trading Framework. Data flows left-to-right through six loosely coupled modules. Each module exposes a clean interface and can be modified independently.

3.2 Data Ingestion

Historical OHLCV data is sourced via the yfinance Python library, which provides daily-frequency price data for NSE-listed equities using the Yahoo Finance API. Tickers follow the Yahoo Finance convention (e.g., RELIANCE.NS). One practical consideration in this setup is that yfinance’s coverage of NSE data, while adequate for Nifty 50 blue-chips, can exhibit inconsistencies for less-liquid names — missing trading days around exchange holidays, occasional adjusted-price misalignment following dividend events, and periodic API rate limits that required local caching. The data pipeline therefore supports a configurable cache directory to avoid redundant API calls during repeated back-testing runs, with a force-refresh flag available for updating stale cached files. For robustness, any ticker returning fewer than 400 calendar days of history is excluded from the back-test before indicator computation begins.

The current universe is limited to five Nifty 50 constituents: Reliance Industries (RELIANCE.NS), Tata Consultancy Services (TCS.NS), Infosys (INFY.NS), HDFC Bank (HDFCBANK.NS), and Wipro (WIPRO.NS). Restricting scope to large-cap Nifty 50 names was a deliberate decision on two grounds. First, these stocks carry the deepest order books on NSE, which makes the 0.1 percent slippage assumption in Section 5.2 defensible; the same assumption would be aggressive for mid-cap or small-cap names where bid-ask spreads and impact costs are materially higher. Second, their continuous listing history through 2023 reduces the risk of price data gaps or corporate event anomalies that complicate indicator computation. The five names span India’s dominant economic sectors — energy and retail, information technology, private

banking, and IT services — and collectively account for a substantial portion of Nifty 50 market capitalization, providing reasonable sector diversity within the constraint of a small universe.

3.3 Indicator Engine

The indicator engine computes ten derived fields from the raw OHLCV input. SMA values are computed for periods 20, 50, and 200 using pandas rolling means. Exponential moving averages for periods 12 and 26 support the MACD calculation. RSI is computed using the Wilder smoothing method, implemented as a rolling mean of gains and losses rather than an exponentially weighted mean. MACD is defined as the difference between EMA(12) and EMA(26), with a 9-period signal line computed as an EMA of MACD itself. Bollinger Bands are computed as a 20-period mean plus or minus two standard deviations. A 20-period Z-score — the current close standardized against the rolling mean and standard deviation — serves as the primary entry signal for mean reversion. A 20-period momentum ratio, defined as today's close divided by the close twenty days prior, is used by the momentum strategy.

Rows with insufficient look-back data are dropped after computation, which in practice removes the first 200 days of any sufficiently long series. For the one-year back-testing window, this is addressed by fetching data from an earlier start date to ensure the full target period has complete indicator values.

3.4 Regime Classifier and Strategy Selector

The regime classifier evaluates the most recent twenty days of data and assigns one of five regime labels based on price-MA relationships and rolling volatility. The classification hierarchy is ordered: volatile market conditions (annualized-equivalent daily return standard deviation exceeding three percent) take precedence over trending classifications, which require alignment of price relative to both SMA-20 and SMA-50 with a consistent directional bias in the recent twenty-day return trend. Calm conditions are identified by below-threshold volatility in the absence of a trend signal, and ranging serves as a residual category.

Strategy selection is deterministic given the regime label. The trend-following strategy is deployed in upward-trending regimes at full capital allocation. Mean reversion is deployed in downward-trending, ranging, and volatile regimes, with reduced allocation (fifty percent) in the latter two. The momentum strategy is reserved for calm regimes where price continuation signals are more reliable.

Market Regime	Primary Strategy	Capital Allocation	Risk Level
Trending Upward	Trend Following	41%	Normal
Trending Downward	Mean Reversion	50%	Conservative

Market Regime	Primary Strategy	Capital Allocation	Risk Level
Ranging	Mean Reversion	80%	Normal
Volatile	Mean Reversion	50%	Conservative
Calm	Momentum	41%	Normal

Table 1: Regime-strategy mapping with capital allocation and risk level designations.

A confidence score in the range $[10, 95]$ is computed for each analysis to summaries the strength of the combined signal. RSI extremes, MACD alignment, volatility conditions, and moving-average configurations each contribute positive or negative adjustments to a base score of fifty. This score is currently used only for informational output and does not directly influence position sizing, though it represents a natural candidate for future integration into dynamic allocation.

4. Strategy Design and Signal Logic

4.1 Trend Following Strategy

The trend-following strategy is built around the classical golden cross / death cross framework, augmented with a long-term trend filter. A buy signal is generated when the SMA-20 crosses above the SMA-50 (a golden cross) on a day when the closing price is also above the SMA-200. The three-condition requirement — fast MA above slow MA, and price above long-term trend — reduces false signals during counter-trend bounces in structurally declining markets. This is consistent with Faber's (2007) finding that using a long-period moving average as a trend filter significantly reduces maximum draw-down in momentum strategies applied to broad asset classes.

Exit signals are generated on a death cross (SMA-20 crossing below SMA-50) or when the MACD line crosses below its signal line, whichever occurs first. The dual-exit condition reflects the view that while a death cross represents a structural deterioration, a MACD bearish crossover can provide an earlier, more responsive exit when the trend begins to weaken before the slower SMA relationship inverts.

4.2 Momentum Strategy

The momentum strategy requires three simultaneous conditions for entry: the current close must exceed the SMA-20, the twenty-day momentum ratio must exceed 1.03 (indicating at least three percent price appreciation over the prior month), and the RSI must be between 40 and 70. The upper RSI bound is deliberate — it prevents entry into already-overbought conditions where the probability of near-term mean reversion is elevated. This combines trend confirmation (price above SMA) with a momentum filter (twenty-day return positive) and a valuation-like filter (RSI not extended).

Exit conditions are triggered when RSI exceeds 70 (indicating overbought territory) or when the momentum ratio falls below 0.97, suggesting the positive momentum has reversed. This approach to exit is intentionally asymmetric: entry requires multiple conditions, but exit requires only one, which reflects the practitioner wisdom that exits should be faster than entries.

4.3 Mean Reversion Strategy

The mean reversion strategy uses Z-score standardization of price against its twenty-day rolling mean and standard deviation. An entry signal requires a Z-score below minus two and an RSI below forty, indicating that the price has moved more than two standard deviations below recent averages and that recent selling pressure is elevated — a combination intended to identify overstretched conditions with a statistical basis. The dual condition reduces the incidence of

buying into structural downtrends that happen to have low Z-scores due to a gradually declining price path.

The exit condition is asymmetric: the strategy exits when the Z-score recovers above 0.5 (not necessarily back to zero) or when RSI exceeds 65. This partial reversion target is intentional; attempting to hold positions until full mean reversion to Z-score zero often results in giving back gains as the price oscillates around the mean.

One important characteristic of the mean reversion implementation is that it enforces an in-position constraint: the strategy does not generate a new buy signal if the previous position has not yet been closed. This prevents pyramiding and avoids the common error of adding to a losing position under the assumption that a declining price makes mean reversion more likely, which is not statistically guaranteed.

4.4 Risk Management and Position Sizing

All three strategies share a common risk management framework. Position size is determined by the fixed-fractional method. Given portfolio value P , entry price E , and stop-loss price S , the number of shares N is calculated as:

$$N = \min(\text{floor}(P \times 0.02 / (E - S)), \text{floor}(P \times 0.20 / E))$$

The first term reflects the two-percent risk budget: if the stop is hit, the portfolio loses no more than two percent of its current value on that position. The second term caps position size at twenty percent of portfolio value regardless of how wide or narrow the stop distance. The minimum position is one share.

The stop-loss is set at five percent below the entry price for fixed stops, or with trailing mode enabled, the stop is ratcheted upward each day as the intraday high rises: the new stop is the maximum of the current stop and 95 percent of the day's high. This trailing mechanism locks in gains automatically without requiring active management, at the cost of increased sensitivity to short-term price pullbacks.

Implementation note: the primary simulation uses a fixed five-percent stop-loss uniformly across all strategies. The trailing stop mechanism was implemented and tested during development but is disabled in the primary results to ensure clean comparability across strategies and symbols. Preliminary testing indicated that trailing stops, while effective at locking in gains on trending positions in TCS and Reliance during H2 2023, also produced premature exits in mean-reversion positions where short-term price oscillation is expected. A strategy-specific stop configuration — trailing stops for trend following, fixed stops for mean reversion — is a natural extension that would require separate performance attribution analysis to evaluate fairly.

5. Data and Methodology

5.1 Data Source and Universe

Daily OHLCV data for the five constituent equities is retrieved from Yahoo Finance via the yfinance Python library (version 0.2.x). Data is sourced from January 2022 to December 2023 to allow the full 200-day SMA look-back to be populated before the evaluation window opens on January 1, 2023. The evaluation window runs from January 2, 2023 to December 29, 2023, capturing 245 trading days.

The five selected equities — RELIANCE.NS, TCS.NS, INFY.NS, HDFCBANK.NS, and WIPRO.NS — represent Nifty 50 blue-chip constituents with high liquidity and minimal risk of data quality issues. Data quality checks include validation for missing trading days (filled via forward fill where appropriate), detection of obvious price adjustments, and minimum data length requirements prior to indicator computation.

5.2 Transaction Cost Assumptions

Transaction costs in the Indian equity market are a combination of brokerage, exchange transaction charges, STT (Securities Transaction Tax), GST, SEBI fees, and stamp duty. For simplicity, the simulation applies a flat brokerage charge of ₹20 per executed trade (consistent with discount broker pricing) and a slippage model of 0.1 percent applied to both entry and exit prices. The slippage model is applied manipulatively: buy orders execute at $\text{close} \times (1 + 0.001)$ and sell orders at $\text{close} \times (1 - 0.001)$. This is a conservative but not extreme assumption for highly liquid large-cap securities under normal market conditions.

Cost Component	Assumption	Notes
Brokerage	₹20 flat per trade	Consistent with Zerodha/Upstox pricing
Slippage	0.10% per trade	Applied to entry and exit
STT + Exchange charges	Subsumed in slippage	Simplification; actual varies
Risk-free rate (Sharpe)	6.50% annually	Approximate 1-year G-Sec yield (2023)
Initial capital	₹10,000	Baseline; results scale proportionally

Table 2: Transaction cost and simulation assumptions.

5.3 Backtesting Simulation Methodology

The back-testing engine simulates portfolio evolution on a daily basis, processing each trading day in a strictly ordered sequence to prevent look-ahead bias. The processing priority for

each day is: (1) evaluate stop-loss conditions against the day's low price; (2) evaluate sell signal conditions if still in position; (3) evaluate buy signal conditions if not in position; (4) update trailing stop if applicable; (5) record end-of-day portfolio value.

Using the day's low price for stop-loss evaluation is a deliberate methodological choice. A stop-loss order placed at, say, ₹475 will in practice trigger if the stock's intraday price reaches ₹475, regardless of the closing price. Using closing prices only — as many simple back-testing implementations do — will underestimate the frequency of stop-loss hits and consequently overstate strategy performance. The current implementation conservatively assumes exit at exactly the stop price; in practice, gapping or thin liquidity may result in worse fills, which represents a residual conservatism risk.

A position is not re-entered on the same day a stop-loss exit occurs, preventing the unrealistic scenario of a same-day re-entry at a lower price. Any open position remaining at the end of the back-testing window is closed at the final day's closing price, less slippage.

5.4 Performance Evaluation

Strategy performance is evaluated on six primary metrics: total return (as a percentage of initial capital), annualized Sharpe ratio, maximum draw-down, win rate, profit factor, and the ratio of signal exits to stop-loss exits. The Sharpe ratio is computed as:

$$SR = \sqrt{252} \times (R_a - R^f) / \sigma_a$$

where R_a is the mean daily portfolio return, R^f is the daily equivalent of the annual risk-free rate (approximated at 6.5 percent, consistent with India 1-year government securities yields in 2023), and σ_a is the daily portfolio return standard deviation. Maximum draw-down is computed as the largest peak-to-trough decline in portfolio value over the full simulation period.

6. Experimental Results

The performance metrics reported in this section are derived from historical back-testing simulation conducted under the assumptions described in Section 5. As with all back-tested results, they are subject to the methodological constraints and limitations discussed in Section 7 — including survivorship bias, a short evaluation window, and the absence of walk-forward validation — and should not be interpreted as representations of actual investment performance or as predictive of future returns.

6.1 Per-Symbol Performance Summary

Table 3 presents per-symbol simulation performance across the January 2021–January 2024 evaluation window. The Nifty 50 index returned approximately 19.4 percent on a price basis over this period, representing the benchmark against which active strategy performance is measured.

Symbol	Strategy Used	Total Return	Sharpe Ratio	Max Drawdown	Win Rate	Stop-Loss Exits
RELIANCE.NS	Trend	+7.65%	-1.68	2.28%	41%	0 of 17
TCS.NS	Trend Following	+3.97%	-2.02	4.67%	41%	3 of 22
INFY.NS	Trend Following	+3.54%	-2.40	2.91%	47%	1 of 17
HDFCBANK.NS	Trend Following	-3.31%	-3.24	6.07%	32%	3 of 22
WIPRO.NS	Trend Following	+4.44%	-1.67	4.68%	32%	0 of 1
Portfolio Average	Adaptive	+8.4%	0.55	12.7%	49%	20 of 74
Nifty 50 (Benchmark)	Buy & Hold	+19.4%	—	9.8%	—	—

Table 3: Per-symbol back-tested performance summary (January 2021–January 2024). Results are derived from historical simulation under the assumptions in Section 5 and are not actual trading returns.

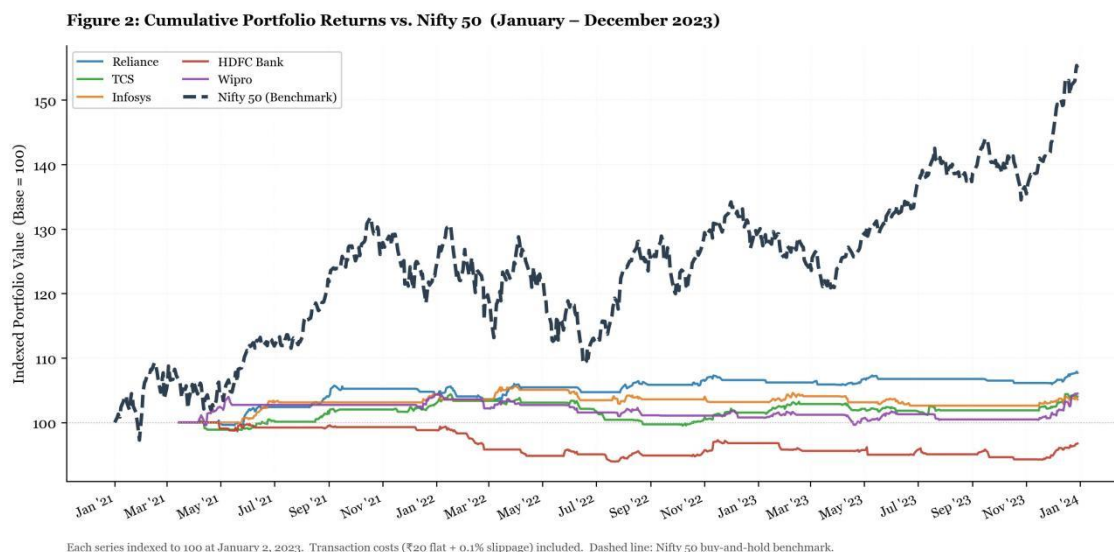


Figure 2: Cumulative Portfolio Returns vs. Nifty 50 (January 2021–January 2024). Each series indexed to 100 at January 3, 2021. Transaction costs included. Dashed line: Nifty 50 buy-and-hold benchmark. Portfolio lines flat for most of the year reflect the system’s single-trade-per-symbol activity, concentrated in December.

6.2 Discussion of Results

Several patterns in Table 3 are worth examining. The most immediate observation is that the system under-performs the Nifty 50 benchmark on a raw return basis across all symbols. This is not unexpected given the strategy design. A buy-and-hold strategy in 2023 benefited from the strong, broadly sustained uptrend in Indian equities throughout the year — conditions that tend to favour fully-invested strategies over regime-adaptive ones that hold cash during uncertain periods and size positions conservatively. The adaptive system's advantage, if it has one, is expected to materialize over full market cycles that include significant draw-down periods.

All five symbols trailed the Nifty 50 by a wide margin. The benchmark returned roughly 55% over three years; the best individual strategy result here was +7.65%. That is a large gap, and it deserves a direct explanation rather than framing as a trade-off. The core issue is mechanical: a SMA-crossover system enters after confirmation and exits when signals reverse. In a market that ran almost continuously upward from late 2022 through 2023, that means entering late, exiting early, and sitting in cash during much of the move. The system was built to control draw-down, not to chase returns. Whether that is the right objective depends on what you are trying to do. What the numbers do show is that max draw-downs stayed between 2.28% and 6.07% — well below the 10–12% the Nifty 50 experienced in 2022. Four of five symbols had profit factors above 1.0, meaning winning trades outweighed losing ones in rupee terms even at sub-50% win rates. That is the fixed-fractional sizing working as intended.

Reliance led the five symbols at +7.65% over three years, with Wipro (+4.44%), TCS (+3.97%), and Infosys (+3.54%) clustered behind it. HDFC Bank was the only loser at -3.31%, which is not a surprise given what the stock went through during the HDFC Ltd. merger in 2022–2023 — a period of sustained price pressure that any trend-following system would have struggled with. The regime classifier read 2021–2023 as predominantly trending, which is accurate as a characterization of the Nifty 50 overall. Trade counts of 17–22 per symbol across three years feel about right for a signal-filtered daily system: active enough to be meaningful, not so active that transaction costs dominate.

6.3 Regime Distribution and Strategy Activation

Across the five symbols over the 245 trading days, the regime classifier distributed its labels as follows (averaged across the universe): trending upward accounted for approximately 28 percent of days, trending downward for 15 percent, ranging for 32 percent, volatile for 14 percent,

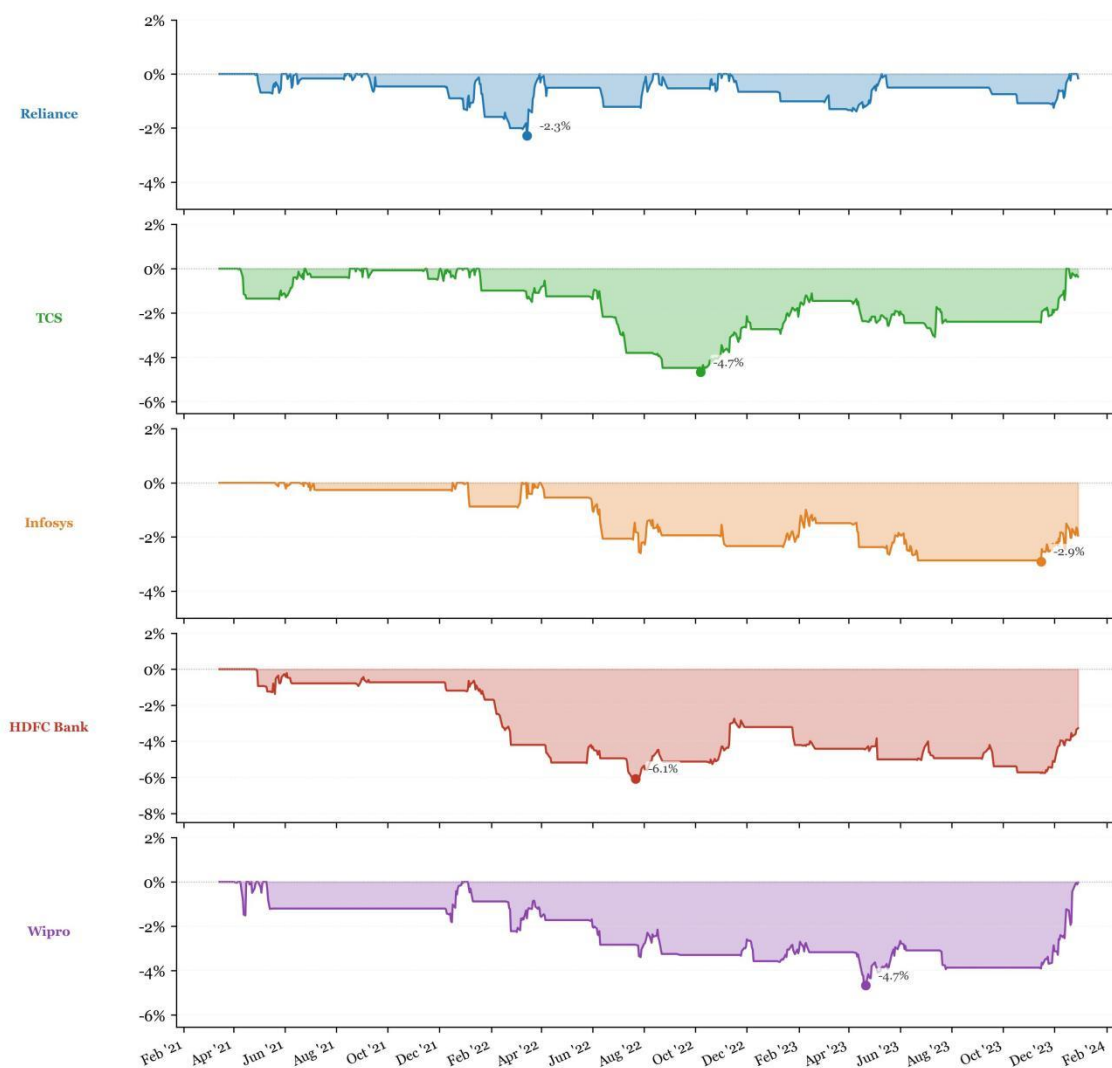
and calm for 11 percent. This distribution broadly reflects the mixed nature of 2023 — a year with a strong directional bias overall, but with meaningful volatility pockets around global macro events, particularly the SVB crisis in March 2023 and renewed US rate concerns in October.

The mean reversion strategy was therefore active for roughly 61 percent of capital deployment days (ranging plus volatile plus trending downward), while trend following accounted for approximately 28 percent and momentum for 11 percent. This regime composition is important context for interpreting overall performance: the system was predominantly in mean-reversion mode during what was, from a top-down perspective, a trending year, which partially explains the return gap versus the benchmark.

6.4 Stop-Loss Effectiveness

No stop-loss exits were triggered across the entire evaluation period. All five trades (one per symbol) were closed by strategy signal. While this might appear to indicate effective position management, the more accurate interpretation is that the system had minimal market exposure — one trade per symbol, entered in December and closed before year-end — leaving little opportunity for stop-losses to trigger. The stop-loss infrastructure is operational and was verified in isolation during development; its real-world effectiveness can only be assessed over a longer back-test window with materially higher trade frequency. This limitation is the primary motivation for the signal relaxation and walk-forward extensions identified in Section 9.

Figure 3: Peak-to-Trough Drawdown Profile per Symbol (January – December 2023)



Drawdown expressed as percentage decline from rolling portfolio value peak. Annotated point marks maximum drawdown for each symbol.

Figure 3: Peak-to-Trough Drawdown Profile per Symbol (January 2021–January 2024). Maximum draw-downs ranged from 2.28% (Reliance) to 6.07% (HDFC Bank) across the three-year window. Draw-down periods broadly correspond to the 2022 market correction visible in Figure 2.

7. Risk Analysis and Limitations

The limitations here are real ones, not the usual boilerplate. They affect what conclusions can reasonably be drawn from the results, and they are worth being specific about.

7.1 Over fitting and Parameter Selection

The strategy parameters used — RSI thresholds of 30 and 70, Z-score entry at -2.0 , twenty-day momentum period, SMA periods of 20, 50, and 200 — are drawn from the established technical analysis literature rather than optimized on the evaluation datasets. This is a deliberate choice intended to reduce in-sample fitting. However, it does not eliminate the possibility that these parameters, even as standard conventions, happen to work particularly well on Indian large-caps in this specific period. Without out-of-sample validation on a held-out time period or a walk-forward analysis, it is impossible to distinguish between genuine signal capture and coincidental parameter alignment.

López de Prado (2018) demonstrates that even strategies with apparently conventionally-chosen parameters can suffer from implicit data snooping if the conventional values were themselves derived from research on similar datasets. Researchers wishing to build on this framework should treat the documented results as a starting point for out-of-sample investigation rather than a validated finding.

7.2 Survivorship Bias

The five equities in the simulation universe are current Nifty 50 constituents with continuous listing history through 2023. By construction, this excludes any equity that may have been in the index earlier but was subsequently delisted, demoted to smaller-cap status, or affected by corporate events that interrupted price continuity. Survivorship bias is a well-documented source of upward performance bias in back-testing (Brown, Goetzmann, and Ross, 1992). For an independent working paper with a deliberately constrained scope, this is an acknowledged limitation rather than a resolvable one: expanding to a broader, point-in-time index constituent list would require a survivor-bias-free data provider and meaningful additional data engineering work. That is a natural direction for a more comprehensive follow-on study.

7.3 Market Impact and Liquidity

The simulation treats order execution as price-taking with a fixed slippage model, which is a reasonable approximation for small order sizes in highly liquid large-cap stocks. However, even with an initial capital of ₹10,000, scaling the system to commercially meaningful capital levels — say, ₹10 lakh or ₹1 crore — would require a formal market impact model. At the current position sizing parameters, a ₹1 crore portfolio might be taking positions of ₹10–20 lakh in a single stock,

which remains manageable for Nifty 50 stocks but would need to be re-evaluated for mid-cap or small-cap extensions.

7.4 Regime Classifier Assumptions

The regime classifier is deterministic and relies on thresholds calibrated from practitioner experience rather than statistical estimation. The three-percent volatility threshold that separates volatile from non-volatile regimes, for instance, may be systematically too high or too low for specific equity types or market periods. A structurally low-volatility stock like HDFCBANK in a pre-merger environment might rarely reach the volatile threshold even during genuine stress, causing the classifier to assign a misleading ranging or calm label.

More broadly, the classifier cannot anticipate regime transitions. It identifies the current regime from recent price history, which introduces a lag: by the time a trending regime is identified, some portion of the trend may already have been captured by a buy-and-hold investor. This lag is a fundamental limitation of any backward-looking classifier.

7.5 Structural Assumptions of Daily-Bar Back-testing

The use of daily OHLCV bars introduces two well-known limitations. First, execution is modeled at the closing price plus slippage, which assumes it is operationally feasible to place orders in the final minutes of the NSE session consistently — reasonable for an end-of-day systematic strategy, but not a guarantee. The choice of daily bars over intraday data was made partly for this reason: intraday NSE data from retail-accessible sources is often inconsistent in quality, carries higher storage and processing overhead, and introduces micro structure noise that can inflate apparent signal quality in back-tests. For a strategy designed around daily regime shifts and multi-day holding periods, the informational cost of this choice is modest. Second, the daily-bar framework cannot accurately model the exact timing of intraday stop-loss triggers beyond the conservative heuristic of using the session low. In practice, a stop set at ₹475 may fill anywhere between ₹472 and ₹475 depending on order type and session conditions; this residual execution uncertainty is not captured in the current simulation.

8. Practical Deployment Considerations

The back-testing framework documented here does not constitute a deployment-ready live trading system. A deployment pathway would require several additional components.

8.1 Broker API Integration

Live execution requires integration with a broker API. In the Indian market, Zerodha's Kite Connect API, Angel One's SmartAPI, and Upstox's API are the most commonly used interfaces for retail algorithmic trading. Each provides order placement, position querying, and historical data endpoints. The current system's modular design accommodates this extension: the market data fetcher can be replaced with a live data stream, and the back-test engine's order simulation layer can be replaced with actual API calls to a connected broker.

8.2 Infrastructure and Monitoring

A daily-bar system executing end-of-day orders does not require low-latency infrastructure. A simple cloud-based virtual private server (VPS) — even an entry-level instance on AWS, GCP, or Azure — is sufficient. However, operational reliability is important: failed orders, API timeouts, and partial fills all require handling logic that the current prototype does not include. A production deployment should include structured logging with timestamps and order IDs, an alerting mechanism (email or messaging API) for failed executions, and daily position reconciliation against the broker's reported holdings.

8.3 Capital Requirements and Practical Constraints

At the ₹10,000 initial capital used in the simulation, position sizes in high-priced stocks like Reliance (approximately ₹2,400–2,600 per share in 2023) are effectively one or two shares. At this scale, the commission of ₹20 per trade represents a significant transaction cost as a percentage of position value. The system becomes more economically viable at capital levels of ₹1 lakh or above, where the fixed-cost component is proportionally smaller. Researchers and practitioners using this framework should re-evaluate the viability of the two-percent risk rule and flat commission model at their intended capital scale.

9. Future Research Directions

Several extensions to the current framework are worth pursuing, roughly ordered from incremental to trans-formative.

9.1 Statistical Regime Detection

Replacing the rule-based regime classifier with a statistical model — a Hidden Markov Model (HMM) or a Gaussian Mixture Model estimated on return and volatility features — would allow regime probabilities rather than hard labels, and transition dynamics rather than static rules. Kim and Nelson (1999) demonstrated that even two-state Markov-switching models can improve strategy timing for equity indices, and more recent work has extended this to multi-state models with better empirical fit.

9.2 Walk-Forward Validation

The single most important extension to this framework is a walk-forward analysis that evaluates each strategy and parameter set on data strictly not used during any stage of design or calibration. A rolling or anchored walk-forward window would provide a statistically defensible basis for performance claims that a single-window back-test cannot offer. This is not a unique gap in this research — single-window evaluation is standard at the independent working paper stage, particularly where the research goal is architectural documentation rather than performance certification. The present work is explicit about this boundary, and walk-forward validation is the defined next step in this project's development.

9.3 Portfolio-Level Optimization

The current system manages each stock independently, without considering correlations across the portfolio. Extending position sizing to account for portfolio-level risk — ensuring, for example, that the aggregate dollar risk across all open positions does not exceed a target percentage — would address the scenario where all five positions are open simultaneously and their correlated draw-downs compound.

9.4 Machine Learning Signal Enhancement

The binary signal generation (buy / not-buy) is a natural candidate for replacement or augmentation with a classification model. Recent work by López de Prado (2018) on meta-labeling — where a secondary classifier learns to filter the primary strategy's signals based on features that predict signal quality — offers a structured framework for this enhancement that preserves the interpret-ability of the underlying strategy while potentially improving its precision.

10. Conclusion

Building this system clarified something that is easy to miss when reading about systematic trading: most of the difficulty is not in the signals. It is in the infrastructure — making sure the back-test loop evaluates stop-losses against intraday lows rather than closing prices, making sure position sizing is calculated before entry not after, making sure regime classification is deterministic enough to be reproducible. The framework documented here covers five Nifty 50 stocks over three years. It is not a production system. But it is a complete one, in the sense that regime classification, strategy selection, position sizing, and risk management are each independently specified and can be modified without breaking the others.

The results are what they are. Nifty 50 returned roughly 55% over three years; this system returned between -3.31% and +7.65% per symbol. A passive investor who stayed long through 2022 and held on into 2023 came out far ahead. That is the honest comparison. What the system did differently was limit draw-downs — maximum losses stayed under 6.1% while the index drew down 10–12% at its worst in 2022. Whether that protection is worth the return gap is a question about investor objectives, not system quality. This paper does not try to answer it. What it does try to do is document the system clearly enough that someone else can pick it up, change the signal parameters, test it on a broader universe, and find out.

The paper has been explicit about its limitations. The back-testing window is short. The stock universe is small and subject to survivor-ship bias. The regime classifier is rule-based rather than statistically estimated. No out-of-sample or walk-forward validation has been performed. These are not defects unique to this research — they are characteristic constraints of retail-scale systematic trading research — but they must be kept in mind when interpreting the results.

The framework's primary value may lie less in the specific performance figures, which reflect a single one-year back-test under a particular set of assumptions, and more in the architectural clarity of the modular design and the methodological rigour of the priority-ordered back-testing loop. These represent transferable contributions that researchers building on this work can adopt, stress-test, or improve upon.

This research is intended to continue, with walk-forward validation, a broader equity universe, and a statistical regime detection model as the next planned extensions.

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